

VCF2

Build Guide

All of the SMD components on the PCB have already been assembled. So, this is a real quick and easy build.

1. Start by placing the 2x5 power header. The placement of the power header is on the downside of the main PCB (the PCB with the square hole and the wavy background is). The header should be soldered from the SMD component side. Solder one pin first and check if the header sits flush with the PCB. If so, you could solder the other pins. If not, reheat the solder joint and adjust the header.
2. Now place and solder the CV_OFFSET1 and the CV_RANGE1 trimpots (mind the orientation, there is a little screw sign printed on the PCB) and cut off the excess legs.
3. Put the RESO_R trimpot into place (mind the orientation again), cut off as much of the protruding legs and solder them afterwards. Try to keep the solder joints as flat as possible. (I would recommend to place a small strip of electrical tape on the solder joints from the topside, to protect the CV jack against issues!)
4. Place the three jack sockets, the three potentiometers and the two vactrols. The placement of each component is marked on the PCB. **Place but do not solder yet!** Mind the orientation of the vactrols! Long leg of the LED into the hole marked with a + sign.
5. Screw on and tighten the knurled nuts from the jacks and the hex nuts from the potentiometers. Have a look if everything is aligned.
6. Now turn the module over and solder the Potentiometers and jack sockets from the SMD Side. Again, cut the protruding legs of the vactrols.

Congratulations, your build is done!

7. Now it is time for the calibration process. CV_OFFSET1 and CV_RANGE1 are 2 trimmers which allow you to adjust the offset and range of the CV signal. So you can scale the controlling CV to fit neatly into the range needed by the vactrols you've used. First, close the filter on the front panel, then adjust the offset until the filter closes, you'll need to trim a fair bit to archive this, and it will happen rather suddenly. Set it where you want the filter to be when fully closed. Then open the filter and adjust the range if needed. Go back and forth between both until you're happy.
8. Depending on how well your home-made or bought vactrols behave, and thus how close the 2 filter poles are together, you might find the filter unable to self-oscillate at full resonance. Turn the RESO_R trimmer till the resonance matches your needs. I like to adjust it, so that the LEDs on the bottom-side of the module does not glow when resonance is fully CCW.